

Metrics reference

Every quantity reported by the colored point-cloud completion benchmark: how it is computed, what it can establish, and — just as important — what it cannot. Written so a number in a W&B table can be read without opening the notebook that produced it.

Colored_Point_Cloud_Completion arms: RePaint-part · joint-6D (PDR) 3 categories × 3 difficulties

1 · What is measured, and where

Every model contributes a ground-truth cloud X of 8192 colored points, split by the occlusion protocol into a **visible** part V (the input) and a **missing** part M . The split is a genuine partition — $V \cup M = X$ exactly — so the target is labelled ground truth, not an approximation.

All color error is measured on M only. This is not a detail. The visible region is handed to the method as input; including it in the average lets a method score well by copying what it was given. On `simple` that would dilute the measurement by a factor of four.

THE CORRESPONDENCE PROBLEM

A method that also generates geometry does not produce a color *at* the ground-truth points — it produces its own cloud G somewhere near them. To score it, each missing ground-truth point $x \in M$ is matched to its nearest generated point $i(x) = \operatorname{argmin}_{g \in G} \|x - g\|$, and that point's color is the prediction.

This makes the color score depend on geometric accuracy. If $i(x)$ sits 10 % of the object's radius away, the color fetched is the color of somewhere else. Section 7 exists to remove this dependency.

2 · Color error: ΔE in CIELAB

RGB distance is not perceptual — the same numeric step is a large visible change in some regions of the cube and invisible in others. Colors are therefore converted to CIELAB, a space built so that Euclidean distance approximates perceived difference, and compared there.

$$\begin{aligned}
 \text{sRGB} \rightarrow \text{linear:} \quad & C_{\text{lin}} = C/12.92 && \text{if } C \leq 0.04045 \\
 & = ((C + 0.055)/1.055)^{2.4} && \text{otherwise} \\
 \text{linear} \rightarrow \text{XYZ:} \quad & [X \ Y \ Z]^T = M \cdot [R \ G \ B]_{\text{lin}} && (\text{sRGB primaries, D65}) \\
 \text{XYZ} \rightarrow \text{Lab:} \quad & t = X/X_n, \ Y/Y_n, \ Z/Z_n && W_n = (0.95047, 1.0, 1.08883) \\
 & f(t) = t^{1/3} && \text{if } t > \delta^3, \ \delta = 6/29 \\
 & = t/(36^2) + 4/29 && \text{otherwise} \\
 & L^* = 116 \cdot f(Y/Y_n) - 16 \\
 & a^* = 500 \cdot (f(X/X_n) - f(Y/Y_n)) \\
 & b^* = 200 \cdot (f(Y/Y_n) - f(Z/Z_n)) \\
 \Delta E \text{ (CIE76)} &= \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2}
 \end{aligned}$$

The reported number is the **mean ΔE over the missing points**. The scale is absolute and carries a physical interpretation:

< 1 Not perceptible to the human eye	1 – 2 Perceptible on close inspection	2 – 10 Perceptible at a glance	10 – 30 Related colors, clearly different	> 30 Effectively unrelated colors
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Commonly cited bands; the classic just-noticeable difference under controlled viewing is $\Delta E \approx 2.3$.

KNOWN LIMITATION

CIE76 is not perceptually uniform everywhere — it overstates differences in saturated blues and understates them in some neutrals. CIEDE2000 corrects this but is not a metric (it violates the triangle inequality), which makes averages and comparisons harder to defend. We report CIE76 for interpretability and state the choice; a paper should say the same rather than imply uniformity it does not have.

3 · Why ΔE structurally favours averaging

ΔE is a **distortion** metric. Like any expected-error measure, it is minimized by the conditional mean of the target given the observation, not by a plausible sample from the posterior:

$$\operatorname{argmin}_{\hat{y}} E[\|\hat{y} - y\|^2 \mid \text{observation}] = E[y \mid \text{observation}]$$

The consequence is structural, not incidental. A method that predicts the average color of what it cannot see will beat a method that commits to one specific, sharp, plausible answer — *even when the second output looks obviously better to a human*. Blau & Michaeli (2018) prove distortion and perceptual quality are in direct tension: improving one past a point must cost the other.

Two of our baselines are pure averaging procedures, and are therefore advantaged by construction:

BASELINE	PROCEDURE	WHY IT IS HARD TO BEAT ON ΔE
NN-copy	Each predicted point takes the color of its nearest <i>visible</i> point	Locally optimal under spatial color smoothness; commits to no new information
part-mean	Each point takes the mean color of its part	Literally the conditional mean, given oracle part labels
global-mean	Every point takes the mean color of the visible region	The coarsest conditional mean; a floor, not a competitor

HOW TO REPORT RESPONSIBLY

A generative method losing on ΔE is *expected* and is not by itself evidence that it is worse. The honest framing pairs ΔE with a distribution-level measure, and reports the posterior mean of the generative method explicitly (Section 7) so the sampling penalty is separated from model quality.

4 · Geometry error: Chamfer distance

$$\begin{aligned} \text{CD}(A, B) = & (1/|A|) \sum_{a \in A} \min_{b \in B} \|a - b\|_2 \\ & + (1/|B|) \sum_{b \in B} \min_{a \in A} \|a - b\|_2 \end{aligned}$$

Symmetric, and L1 in the sense that distances enter unsquared — squared Chamfer lets a handful of outliers dominate. Both terms matter and mean different things: the first penalizes predicted points that sit nowhere near the surface (*precision*), the second penalizes surface left uncovered (*recall*).

Clouds are centred and divided by their maximum radius before scoring, so the object fits a unit ball and Chamfer reads as a fraction of object size: **CD = 0.05 means roughly 5 % of the object's radius**. This makes values comparable across categories.

The ratio to doing nothing

```
chamfer_vs_partial_x = CD(partial, GT) / CD(generated, GT)
```

Raw Chamfer answers "how good", this answers "was it worth running the model". Above 1, completion beats handing back the input untouched; at 1 the model has added nothing; below 1 it has made things worse. The denominator flips it so higher is better, matching every other "x" column.

READ THIS ONE CAREFULLY

The partial cloud is a *subset of the ground truth*: every point it contains is exactly right. Its Chamfer is non-zero only through the recall term. So the ratio compares a high-precision / low-coverage cloud against a full-coverage / lower-precision one, and it is expected to sit near 1 when little is occluded. That is a property of the comparison, not a failure of the model.

5 · Difficulty

Occlusion follows the PoinTr / ShapeNet-55 protocol: draw a random unit viewpoint v , sort all points by distance to v , and remove the nearest $r \cdot N$. The seed is recorded, so every partial is reproducible from the ground truth alone.

LEVEL	CROP RATIO R	VISIBLE	WHAT IT PROBES
simple	0.25	75 %	Refinement — most of the object is given
moderate	0.50	50 %	Balanced completion
hard	0.75	25 %	Genuine generation from a fragment

EMPIRICAL FINDING — THIS AXIS DOES NOT CONTROL COLOR DIFFICULTY

Across a category, moving from 25 % to 75 % occlusion roughly doubles Chamfer but moves ΔE by only about 2 units. Across categories, ΔE varies threefold (chair 12–18, airplane 21–27, car 30–38). Color difficulty is set by **texture complexity**, not by how much is hidden. A colored-completion benchmark that inherits difficulty levels from a geometry benchmark is therefore mislabelling its own axis — a second, color-specific axis is needed.

6 · Metric catalogue — main run

Logged per model to `joint6d/<category>/<difficulty>`, aggregated into `SUMMARY-joint6d`.

COLUMN	BETTER	DEFINITION
chamfer_gen	lower	CD(generated, GT) — geometric error of the produced cloud
chamfer_partial	–	CD(partial, GT) — reference, the cost of returning the input unchanged
chamfer_vs_partial_x	higher	Ratio of the two; > 1 means completion earned its keep
dE_joint6d	lower	Mean ΔE on M, prediction fetched through the nearest-generated-point map $i(x)$
dE_nn_on_gen	lower	Same map, but colors come from NN-copy instead of the model. Same geometry, same correspondence — the difference isolates coloring

Pairing `dE_joint6d` with `dE_nn_on_gen` rather than an absolute threshold is deliberate: both travel through the identical geometry and the identical nearest-neighbour map, so whatever error those introduce is common to both and cancels in the comparison.

7 · Metric catalogue — ablation

The main-run gap conflates three separate effects. The ablation notebook trains nothing; it reloads the checkpoints and separates them.

free — the main-run setup, with K samples

Geometry and color both generated. $K = 5$ samples per model.

anchor — geometry pinned to ground truth

At every diffusion step the xyz channel is replaced by the correctly-noised ground-truth geometry, so only color evolves freely — the channel-wise form of RePaint's replacement trick:

```
for t = T-1 ... 0:
    x ← [ q_sample(X_gt, t) || x_rgb ]      # xyz pinned at noise level t
    x ← p_sample( model(x, t, cond), x, t )
```

Generated color then lands *exactly on* the ground-truth points, $i(x)$ becomes the identity, and the correspondence drops out of the measurement entirely. What remains is pure color error. The model was not trained this way — it is an inference-time diagnostic, not a claim about the trained model.

COLUMN	DEFINITION
dE_free_sample	Mean over K samples of the per-sample ΔE . Reproduces the main-run metric
dE_free_postmean	ΔE of the <i>color-space average</i> of the K samples — the posterior-mean estimate ΔE actually rewards
dE_free_nn	NN-copy on the generated geometry (main-run baseline, recomputed as a control)
dE_anchor_sample	Per-sample ΔE with geometry pinned to GT
dE_anchor_postmean	K-sample color average, geometry pinned to GT — the cleanest measure of the model's coloring
dE_anchor_nn	NN-copy evaluated on GT geometry — the baseline on identical footing
dE_anchor_global	Every missing point painted the mean color of the visible region — the prior floor

8 · Derived quantities, and how to read them

QUANTITY	DEFINITION	READING
sampling_penalty	$\text{free_sample} - \text{free_postmean}$	Positive → part of the gap is the variance cost of drawing one sample, not model quality. Large values mean ΔE is penalizing the act of sampling
geometry_penalty	$\text{free_sample} - \text{anchor_sample}$	Positive → the nearest-neighbour correspondence was inflating the color score. This is measurement error, not coloring error
gap_free	$\text{free_sample} - \text{free_nn}$	The main-run headline, recomputed. Serves as a consistency check against the original run
gap_anchored	$\text{anchor_postmean} - \text{anchor_nn}$	The decisive number. Both sides on ground-truth geometry, both at their best form. Negative means the generative model genuinely beats NN-copy

QUANTITY	DEFINITION	READING
prior_margin	$\text{anchor_global} - \text{anchor_postmean}$	Positive → the model beats a constant color, so conditioning is doing something. Near zero → the color head is emitting a category prior and ignoring the visible colors

THE DECOMPOSITION

Read together: gap_free is what was published; sampling_penalty and geometry_penalty say how much of it was method artefact; gap_anchored is what survives once both are removed. If gap_anchored stays positive, the negative result is real and reportable. If it goes negative, the original conclusion was an artefact of single-sample evaluation through a noisy correspondence — a materially different paper.

9 • Reference values

Numbers to anchor any new result against. All are ΔE unless noted.

QUANTITY	VALUE	WHAT IT ESTABLISHES
Within-part ΔE , real texture	20.7	Structural ceiling for any part-constant coloring on real ShapeNet texture — no such method can score below this
Within-part ΔE , synthetic texture	5.5	The same ceiling on flat synthetic color; the 4× gap is why Stage-1 real texture was necessary
Textured fraction of models	74 %	Share of models with genuine texture (4 % flat) — the benchmark is not dominated by trivial cases
Joint-6D vs NN-copy, 9 cells	+4.06 ± 0.94	Systematic, low-variance loss across every category and difficulty — unlike the RePaint arm, which wins at the two lower levels
RePaint-part vs NN-copy, airplane simple / moderate / hard	−4.30 −3.98 +2.99	Wins at two of three levels. Supersedes a pre-fix figure of +9.61 measured before the density shift and frame-verdict bugs were corrected
Part conditioning vs part-blind, airplane	−4.9 ... −7.0	RePaint-part against RePaint-vanilla, consistent across all three levels — the gain comes from part awareness, not from diffusion alone
RePaint-part vs oracle ceiling, airplane/simple	−0.47	Beats the part-constant structural ceiling, so the model produces correct <i>within-part</i> variation rather than a per-part mean
Oracle part labels, airplane	≤ 1.44	Most that perfect segmentation buys over the trained segmenter — segmentation is not the bottleneck
Chamfer floor, 2048 points	≈ 0.05	Best geometric precision observed at this resolution, independent of difficulty

10 • What these metrics cannot tell you

- **Whether a result looks right.** ΔE is a per-point average and is blind to spatial structure — a prediction with correct colors in wrong places can score identically to one with the arrangement right. No distribution- or structure-level metric is currently reported.
- **Whether the method is better for a human viewer.** Distortion and perceptual quality are provably in tension (Section 3). Winning on ΔE is not winning on appearance, and losing on it is not losing on appearance.
- **Anything about categories outside the three benchmarked.** ΔE level is governed by texture complexity, which varies threefold within these three alone. Extrapolation is unjustified.

- **Statistical significance.** Aggregates are means over 15 test models per cell. Spreads are reported where they exist; no significance test is run, and none should be implied.

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`Joint6D_PDR_benchmark.ipynb` and `Joint6D_ablation.ipynb`